

Proposal: Solar + Battery Microgrid for Chicken Republic (CR_Ilupeju)

Prepared for: Chicken Republic — CR_Ilupeju (branch_id: 83)

Objective: Reduce grid + diesel spend, improve uptime, and stabilize power quality by placing the entire branch load behind a hybrid inverter/PCS that blends **Solar + Battery + Grid + Generator**.

1) Executive Summary (The Offer)

We propose a **whole-site, inverter-backed microgrid** for CR_Ilupeju built around a **100 kW hybrid PCS**.

Two options (same architecture, different PV size):

- **Option 1 — Roof PV (base): ~40.3 kWp** (65 × 620 W panels)
- **Option 2 — Roof PV + Carport PV (upgrade): ~65.1 kWp** (105 × 620 W panels total)

Financial headline (using your energy costs):

- Energy prices used: **Grid ₦230/kWh, Generator ₦309/kWh**
- Indicative payback (using provided component CAPEX only):
 - **Without carport: ~3.26 years**
 - **With carport: ~2.57 years**

2) What the Data Says (GEN + UTILITY only)

From the last 12 months of CR_Ilupeju readings:

- **Annual energy: ~215,634 kWh/year** ($\approx$ 606 kWh/day)
- **Average load (energy-based): ~25 kW**
- **Peak observed demand: ~83 kW**
- **Supply mix:**

- **Grid (UTILITY): ~183,496 kWh/year**
- **Generator: ~32,138 kWh/year** (generator operated on ~250 days)

This profile is ideal for solar because daytime demand is strong (little PV wastage) and generator usage is material (diesel savings upside).

3) System Architecture (Whole-load Through the Inverter)

Single protected bus: all branch loads are fed from the hybrid PCS output.

How it works:

- Solar serves load first; surplus charges battery
- Battery supports load during dips/changeovers
- PCS imports from grid when needed; when grid is unavailable, PCS accepts/starts generator
- PCS controls blending to avoid reverse power into generator and smooth transients

Must-have electrical scope: maintenance bypass, protection coordination, proper generator interface (anti-backfeed / reverse power), metering + monitoring.

4) Recommended Sizing

4.1 Hybrid inverter/PCS

- **100 kW 3-phase hybrid PCS** (whole-load rating)
 - Sized above the measured **~83 kW** peak with margin

4.2 Battery (recommended)

- **60 kWh LFP** (best balance of ROI + operational value)
 - Seamless transfer support, stability for cold room loads, generator smoothing

4.3 Solar PV (two options)

- **Option 1 (no carport):** 65 panels $\times$ 620 W = **40.3 kWp**
- **Option 2 (with carport):** (65 + 40) panels $\times$ 620 W = **65.1 kWp**

5) Expected Energy & Savings

5.1 PV production assumption

- Lagos yield model: **PSH = 5.0**, **PR = 0.75**
- Expected yield $\approx$ **1,369 kWh/kWp/year**

5.2 Expected PV energy

- **Option 1 (40.3 kWp):** $\sim$ 55,171 kWh/year
- **Option 2 (65.1 kWp):** $\sim$ 89,122 kWh/year

5.3 Value of energy avoided

We value each solar kWh using a blended avoided-cost basis reflecting daytime grid/gen usage:

- **Blended avoided cost $\approx$ ₦244/kWh** (grid ₦230, generator ₦309)

Estimated annual savings:

- **Option 1:** $55,171 \times 244 \approx$ **₦13.46m/year**
- **Option 2:** $89,122 \times 244 \approx$ **₦21.75m/year**

6) ROI / Payback (Using Your CAPEX Benchmarks)

Note: CAPEX below includes **PCS + battery + panels (+ carport)** as provided.
BOS/installation/protection can be added after site survey for an “all-in” ROI.

6.1 CAPEX

Item	Without Carport	With Carport
Hybrid PCS (100 kW)	₦15.00m	₦15.00m
Battery (60 kWh LFP)	₦19.20m	₦19.20m
Panels (₦150k each)	65 panels = ₦9.75m	105 panels = ₦15.75m
Carport structure	—	₦6.00m
Total CAPEX	₦43.95m	₦55.95m

6.2 Payback

- **Without carport:** $43.95 / 13.46 \approx 3.26$ years
- **With carport:** $55.95 / 21.75 \approx 2.57$ years

6.3 Carport upgrade economics (incremental)

- Incremental cost (carport + 40 panels): **₦12.0m**
- Incremental savings: $(89,122 - 55,171) \times 244 \approx \text{₦8.28m/year}$
- **Incremental payback:** $12.0 / 8.28 \approx 1.45$ years

7) What CR_Ilupeju Gets (Beyond Savings)

- **Lower diesel exposure:** PV continues producing even when the branch is on generator (blended operation)
- **Higher uptime + better power quality:** inverter-backed bus stabilizes voltage/frequency for sensitive loads
- **Reduced generator wear:** fewer starts, better loading, lower maintenance burden
- **Remote monitoring & reporting:** visibility for performance, alarms, and energy KPIs

8) Key Assumptions (For This Financial Model)

- Energy prices: **Grid ₪230/kWh, Generator ₪309/kWh**
- PV yield basis: **PSH 5.0, PR 0.75** (~1,369 kWh/kWp/year)
- Solar energy is assumed to be largely self-consumed due to strong daytime demand
- CAPEX table includes only items explicitly provided (PCS, battery, panels, and carport)

9) Next Steps

1. Site survey (DB capacity, protection, earthing, space confirmation, cable routes)
2. Confirm coupling approach for PV expansion (DC-coupled vs AC-coupled) and finalize SLD
3. Finalize BOS + installation cost and lock the “all-in” ROI and delivery schedule

We recommend proceeding with Option 1 immediately, and treating the carport PV as a high-ROI upgrade path (≈1.45-year incremental payback).